A remark on depth-three circuits and a homogeneous depth-five lower bound for iterated matrix multiplication

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Abstract

We study the size of depth-three ($\Sigma\Pi\Sigma$) circuits computing the iterated matrix multiplication polynomial $\mathrm{IMM}_{n,n}$ over fields of characteristic zero. The result is conditional on a lower bound for homogeneous depth-five circuits stated in a recent unrefereed manuscript [1]. Assuming that bound, $\Sigma\Pi\Sigma$ circuits computing $\mathrm{IMM}_{n,n}$ have size at least $n^{(1/400-o(1))\sqrt{n}}$, and $\Sigma\Pi\Sigma$ circuits computing the $M \times M$ determinant or permanent have size $M^{\Omega(M^{1/4})}$. The bound for $\mathrm{IMM}_{n,n}$ matches the known upper bound $n^{O(\sqrt{n})}$ up to the constant in the exponent, whereas the best unconditional lower bound known to us is $n^{\Omega(\log n)}$. The reduction from depth three to homogeneous depth five is known; the note adds its combination with the manuscript's bound at degree n, a vertex count for the resulting circuit in the manuscript's model, and the corollary for the determinant and the permanent. The bounds are asymptotic, and the bound for $\mathrm{IMM}_{n,n}$ is non-trivial only for $n > e^{1600}$.

1 Introduction

For $n \geq 1$ let $X^{(1)}, \ldots, X^{(n)}$ be $n \times n$ matrices whose entries are distinct commuting variables and let $\mathrm{IMM}_{n,n} = (X^{(1)} \cdots X^{(n)})_{1,1}$, a homogeneous polynomial of degree n in n^3 variables. This note concerns the size of depth-three ($\Sigma\Pi\Sigma$) circuits, that is, sums of products of affine forms, computing $\mathrm{IMM}_{n,n}$ over a field of characteristic zero. The size of a $\Sigma\Pi\Sigma$ circuit is measured here by the number of its product gates plus the number of its distinct bottom affine forms; the definitions are in Section 2. By the depth-reduction theorem of Gupta, Kamath, Kayal and Saptharishi [3], $\mathrm{IMM}_{n,n}$ has $\Sigma\Pi\Sigma$ circuits of size $n^{O(\sqrt{n})}$ (Proposition 6). In the other direction, the results of Amireddy, Garg, Kayal, Saha and Thankey [4] give the unconditional lower bound $n^{\Omega(\log n)}$ on the number of nodes of such a circuit, as explained under "Previous bounds" below.

The result. The result depends on the following statement, quoted verbatim from [1].

Hypothesis A ([1, Corollary 6.1])**.** *"There is an absolute integer n_0 such that, for every field K of characteristic zero and every $n \geq n_0$, every syntactically homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit over K computing $\mathrm{IMM}_{n,n}$ has at least $n^{\sqrt{n}/400}$ gates. Bottom fan-in and bottom support are unrestricted."*

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Assuming Hypothesis A, every $\Sigma\Pi\Sigma$ circuit computing $\mathrm{IMM}_{n,n}$ over a field of characteristic zero has size at least $n^{\sqrt{n}/400-4\sqrt{n}/\ln n} = n^{(1/400-o(1))\sqrt{n}}$ for all sufficiently large n (Theorem 3), and every $\Sigma\Pi\Sigma$ circuit computing the $M \times M$ determinant or permanent over such a field has size $M^{\Omega(M^{1/4})}$ (Corollary 5). The lower bound for $\mathrm{IMM}_{n,n}$ matches the upper bound $n^{O(\sqrt{n})}$ up to the constant in the exponent (see the end of Section 4). Table 1 lists the bounds.

Polynomial	Size of $\Sigma\Pi\Sigma$ circuits	Source	Assumption
$\mathrm{IMM}_{n,n}$	at most $n^{O(\sqrt{n})}$	[3, Theorem 1.1]; Proposition 6	none
$\mathrm{IMM}_{n,n}$	at least $n^{\Omega(\log n)}$	from [4, Corollary 4.13]	none
$\mathrm{IMM}_{n,n}$	at least $n^{(1/400-o(1))\sqrt{n}}$	Theorem 3	Hypothesis A
$\det_M$	at most $M^{O(\sqrt{M})}$	[3]	none
$\det_M$, per_M	at least $M^{\Omega(M^{1/4})}$	Corollary 5	Hypothesis A

Table 1: Bounds on the size of $\Sigma\Pi\Sigma$ circuits over fields of characteristic zero. The unconditional lower bound is for the number of nodes, the measure of [4]; the conditional ones are for the smaller quantity $s + B$.

Ingredients and attribution. The reduction from depth three to homogeneous depth five is known. It is [2, Lemma 10]: over a field of characteristic zero (or greater than d), a $\Sigma\Pi\Sigma$ circuit of size s computing a homogeneous polynomial of degree d can be converted to a homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit of size $\mathrm{poly}(s)\exp(O(\sqrt{d}))$. It rests on the expression of elementary symmetric polynomials through power sums by Newton's identities, which Shpilka and Wigderson [5, Section 4.5.1] used to construct small formulas for the elementary symmetric polynomials; see [6, Lemma 23.6] for an exposition.[1] The route from depth three through homogeneous depth five is not new either. Kayal and Saha [9] state the reduction as their Lemma 5 ("implicit in" the conference version of [3]), observe that a $\Sigma\Pi\Sigma$ circuit of size $N^{o(\sqrt{d})}$ yields a homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ formula of size $N^{o(\sqrt{d})}$ and that the reduction preserves the bottom fan-in, and use it to obtain lower bounds for depth-three circuits of small bottom fan-in over fields of characteristic zero by way of a subclass of homogeneous depth-five circuits; they ask whether the restriction of small bottom fan-in can be dropped from both models. Limaye, Srinivasan and Tavenas [2] obtain the product-depth-one case of their Corollary 4, for $d \le (\log n)/100$ and with no restriction on the bottom fan-in, by a route that begins with the same reduction: they homogenise the circuit, pass to a set-multilinear circuit, and prove the lower bound there. The depth-five lower bound is due to [1]; it has no restriction on the bottom fan-in. What is added here is the observation that the two combine at degree $d = n$, a verification that the homogenised circuit lies in the exact model of [1] together with a count of its vertices (Lemma 1), and the corollary for the determinant and the permanent. The note contains no new lower-bound technique: Theorem 3 is a consequence of Hypothesis A, not an independent lower bound.

Conditional status. Theorem 3 and Corollary 5 are conditional on Hypothesis A, and the manuscript [1] that states it has not been refereed. Its lower bound has two parts: an upper

[1] In [2] the lemma carries the attribution "Lemma 5.6 in the journal version" of [3]. In the report version of [3] cited here, Lemma 5.6 converts a $\Sigma\Pi\Sigma$ circuit with top fan-in s and product gates of fan-in at most d, computing a polynomial of degree n in N variables, into a depth-five powering $(\Sigma\wedge\Sigma\wedge\Sigma)$ circuit of size $\mathrm{poly}(N, d, s)\, 2^{O(\sqrt{n}\log n)}$; the statement used here is [2, Lemma 10].

bound for a rank measure on circuits (its Proposition 3.1) and a lower bound for the same measure on $\mathrm{IMM}_{n,n}$ (its Proposition 4.1). We did not independently verify Proposition 4.1, and we have not checked the Lean formalisation of the bound that accompanies the manuscript. The lemmas, Proposition 6 and the bound $n^{\Omega(\log n)}$ do not depend on Hypothesis A.

Previous bounds. Shpilka and Wigderson [5] proved near-quadratic lower bounds for depth-three formulas over fields of characteristic zero, and Kayal, Saha and Tavenas [7] proved an $\Omega(n^3/(\ln n)^2)$ lower bound for depth-three circuits computing an explicit polynomial in n variables over any field. Limaye, Srinivasan and Tavenas [2] proved the first superpolynomial lower bounds for constant-depth circuits over fields of characteristic zero; their Corollary 4 shows that for $d \le (\log n)/100$ every circuit of product-depth one (at most one product gate on every path from the output to a leaf; this includes $\Sigma\Pi\Sigma$ circuits) computing $\mathrm{IMM}_{n,d}$ has size $n^{\Omega(\sqrt{d})}$, which they note is tight by [3]. Amireddy, Garg, Kayal, Saha and Thankey [4] removed most of the restriction on the degree: specialised to product-depth one, [4, Corollary 4.13] states that for $n = \omega(d)$ every circuit of product-depth one computing $\mathrm{IMM}_{n,d}$ over a field of characteristic zero or greater than d has size at least $2^{-O(d)} n^{\Omega(\sqrt{d})}$. Taking $d = \lfloor \varepsilon \log^2 n \rfloor$ with a small constant $\varepsilon > 0$ gives $n^{\Omega(\log n)}$, and $\mathrm{IMM}_{n,d}$ is obtained from $\mathrm{IMM}_{n,n}$ by substituting identity matrices for $n - d$ of the factors. Both [2] and [4] measure the size of a circuit by the number of its nodes (in [4], of gates, the leaves being counted as input gates). A circuit of the form (1) for $\mathrm{IMM}_{n,d}$, with the variables outside the matrix entries set to zero, has at most $s + B + n^3 + 2$ nodes (the output, s product gates, B bottom sums, at most n^3 variables and the constant 1), so for circuits in which every multiplicity m_{ij} equals one their lower bounds pass to $s + B$ up to this additive term. For circuits with larger multiplicities this would require that a product gate may read the same node m_{ij} times, a point on which neither source is explicit; we therefore state the unconditional bound in the measure of [2, 4]: depth-three circuits computing $\mathrm{IMM}_{n,n}$ over characteristic zero have $n^{\Omega(\log n)}$ nodes. We are not aware of a stronger unconditional bound for $\mathrm{IMM}_{n,n}$ in the literature. The measure $s + B$ of Theorem 3 is at most the number of nodes under either convention, so the comparison is in the direction that matters: Theorem 3 would replace $\log n$ by $(1/400 - o(1))\sqrt{n}$ in the exponent.

For homogeneous depth-five circuits, the bounds of [2, 4] have no restriction on the bottom fan-in, but they are superpolynomial only for small degree (for $d \le \varepsilon \log^2 n$ in [4], which reports a smaller range for [2]), where they give $n^{\Omega(\log n)}$ at best. The earlier bounds known to us at degree polynomial in the number of variables restrict the bottom fan-in or are over finite fields. Kayal and Saha [9, Theorem 2] prove the lower bound $N^{\Omega(\sqrt{d})}$ for homogeneous depth-five formulas of bottom fan-in $O(N^\mu)$, for any fixed $\mu < 1$, computing an explicit N-variate polynomial of degree d in VNP over a field of characteristic zero. Bera and Chakrabarti [10, Theorem 1.1] prove that for every constant $\mu \in [0, 1/2)$ there is an integer q such that every homogeneous depth-five circuit of bottom fan-in N^μ computing the iterated product of n matrices of order n^q, a polynomial of degree n in N variables, has size $N^{\Omega(\sqrt{n})}$, over any field. Over finite fields, exponential lower bounds without such a restriction were proved by Kumar and Saptharishi [8], who also discuss the difficulty of the characteristic-zero case. Hypothesis A differs from the bounds of [9] and [10] in having no restriction on the bottom fan-in, and it is this that lets the reduction from general depth-three circuits apply.

We are not aware of a statement of the consequence recorded here. The only follow-up to [1] known to us is an unrefereed remark [11], which claims that a modification of the proof in [1] gives, for the same homogeneous depth-five circuits, the lower bound $n^{(1/(4\sqrt{2})-\varepsilon)\sqrt{n}}$ for $n \ge n_0(\varepsilon)$, with explicit thresholds; it does not consider depth-three circuits, the determinant or the permanent, and it is not used here (Section 5).

2 Circuit models

The model of [1]. A circuit is a finite directed acyclic graph whose leaves are variables or field elements; a sum gate computes a linear combination of its inputs with arbitrary scalar coefficients, and a product gate computes the product of its inputs, where a gate may occur repeatedly among the inputs and repeated occurrences are counted with multiplicity. A $\Sigma\Pi\Sigma\Pi\Sigma$ circuit has five layers of gates, of types $+, \times, +, \times, +$ from the output to the leaves, a single output gate, edges only between consecutive layers, and bottom sums reading leaves; unary gates are allowed and fan-in, fan-out and sharing are unrestricted. It is *syntactically homogeneous* if, with variables of formal degree 1, constants of formal degree 0 and product gates adding the degrees of their inputs with multiplicity, all inputs of every sum gate have the same formal degree, which is then the formal degree of that gate; an empty sum has formal degree 0. The size is the number of vertices, including leaves; this is the quantity bounded in [1, Theorem 1.1].[2] The manuscript also defines the number of computational gates, which excludes leaves, and says that its lower bound is on gates. We use Hypothesis A only in the weaker form in which "gates" means vertices, leaves included; the count in Lemma 1 below bounds the number of vertices.

Depth-three circuits. A $\Sigma\Pi\Sigma$ circuit computes

$$f = \sum_{i=1}^{s} c_i \prod_{j \in J_i} L_j^{m_{ij}}, \tag{1}$$

where $L_1, \dots, L_B$ are the distinct affine forms feeding product gates, $J_i \subseteq \{1, \dots, B\}$, $m_{ij} \geq 1$ and $c_i \in K$. Thus s is the number of product gates and B the number of distinct bottom forms. We call $s + B$ the *size* of the $\Sigma\Pi\Sigma$ circuit; it is at most the number of gates. Neither s nor B depends on the multiplicities m_{ij}, and the product gates may have degree far larger than $\deg f$.

3 Homogenisation with a vertex count

Write $L_j = a_j + \ell_j$ in (1), with $a_j \in K$ and ℓ_j a homogeneous linear form. Let $p(n)$ denote the number of partitions of n; it is nondecreasing in n. For a partition λ let $\mathrm{len}(\lambda)$ be its number of parts and $z_\lambda = \prod_r r^{\mu_r} \mu_r!$, where μ_r is the number of parts of λ equal to r. Power sums are written P_t. The following lemma is [2, Lemma 10] with the circuit placed in the model of [1] and its vertices counted. In it the set of V variables is fixed in advance and may contain variables on which f does not depend; all V of them are leaves of the circuit constructed and are counted. In the lemma n denotes the degree of the component computed; in Theorem 3 it is also the parameter of $\mathrm{IMM}_{n,n}$.

Lemma 1. *Let* $\mathrm{char}\, K = 0$ *and let* f *be as in* (1)*, in* V *variables. For every* $n \geq 1$ *the homogeneous component* f_n *of degree* n *of* f *is computed by a syntactically homogeneous* $\Sigma\Pi\Sigma\Pi\Sigma$ *circuit over the same* V *variables with at most* $V + 2B + nB + ns + s\,p(n) + 1$ *vertices.*

Proof. Normalisation. A term of (1) in which some L_j, $j \in J_i$, is the zero form vanishes; discard it. If $L_j = a_j \neq 0$ is a nonzero constant ($\ell_j = 0$), remove j from J_i and replace c_i by $c_i a_j^{m_{ij}}$. This changes neither f nor the bounds s, B, and afterwards $\ell_j \neq 0$ for every j that occurs. Put $J_i^0 = \{j \in J_i : a_j = 0\}$, $J_i^1 = J_i \setminus J_i^0$ and $D_0(i) = \sum_{j \in J_i^0} m_{ij}$. Then

$$\prod_{j \in J_i} L_j^{m_{ij}} = \Big(\prod_{j \in J_i^1} a_j^{m_{ij}} \Big) \Big(\prod_{j \in J_i^0} \ell_j^{m_{ij}} \Big) \prod_{j \in J_i^1} \Big(1 + \frac{\ell_j}{a_j} \Big)^{m_{ij}}.$$

[2] The formal statement accompanying [1] uses the same count: the sum of the numbers of leaves and of gates in the five layers.

The middle factor is homogeneous of degree $D_0(i)$. If $D_0(i) > n$, or if $J_i^1 = \emptyset$ and $D_0(i) \neq n$, the term has no component of degree n; discard it. For the remaining terms let $k_i = n - D_0(i) \geq 0$ and let y_i be the multiset of linear forms containing ℓ_j/a_j with multiplicity m_{ij}, $j \in J_i^1$. The component of degree n of the ith term is

$$c_i\Big(\prod_{j\in J_i^1} a_j^{m_{ij}}\Big)\Big(\prod_{j\in J_i^0} \ell_j^{m_{ij}}\Big)e_{k_i}(y_i),$$

where e_k is the elementary symmetric polynomial and $e_0 = 1$.

Newton's identities. Comparing coefficients of z^k in $\prod_y(1+yz) = \exp\big(\sum_{t\geq 1}(-1)^{t-1}P_t z^t/t\big)$, which is valid over any $\mathbb{Q}$-algebra, gives

$$e_k(y_i) = \sum_{\lambda\vdash k} \frac{(-1)^{k-\mathrm{len}(\lambda)}}{z_\lambda} \prod_r P_r(y_i)^{\mu_r}, \qquad P_t(y_i) = \sum_{j\in J_i^1} m_{ij} a_j^{-t}\, \ell_j^t.$$

The multiplicities enter only through the scalars $m_{ij}a_j^{-t}$.

The circuit. Its leaves are the V variables; no constant leaves are used.

- Bottom sums: one gate computing ℓ_j for each j that occurs; at most B gates. Each reads at least one variable leaf, since $\ell_j \neq 0$.
- Lower products: ℓ_j^t for $1 \leq t \leq n$, the bottom gate ℓ_j taken t times; at most nB gates, shared between the terms.
- Middle sums: for each remaining i with $J_i^1 \neq \emptyset$ and each $1 \leq t \leq n$, a gate computing $P_t(y_i)$ as a linear combination of the lower products ℓ_j^t, $j \in J_i^1$ (at most ns gates, each with at least one input); and for each j a unary gate U_j reading the lower product ℓ_j^1 (at most B gates).
- Upper products: for each remaining i and each partition λ of k_i, the product of the gates U_j, $j \in J_i^0$, each taken m_{ij} times, and of the gates $P_r(y_i)$, each taken μ_r times. There are $p(k_i) \leq p(n)$ of them for each i, hence at most $s\,p(n)$. (If $k_i = 0$ the only partition is the empty one and no gate P_r is used; if $k_i > 0$ then $J_i^1 \neq \emptyset$, so the gates $P_r(y_i)$ exist.)
- Output: the linear combination of the upper products in which the product for (i,λ) has coefficient

$$c_i\Big(\prod_{j\in J_i^1} a_j^{m_{ij}}\Big)\frac{(-1)^{k_i-\mathrm{len}(\lambda)}}{z_\lambda}.$$

Edges join consecutive layers only and there is one output gate. The formal degrees are: bottom sums 1; the lower product ℓ_j^t has degree t; the middle sums $P_t(y_i)$ have degree t and the gates U_j degree 1; every upper product has degree $D_0(i) + \sum_r r\mu_r = D_0(i) + k_i = n$; the output has degree n. Every sum gate except possibly the output has at least one input, and all inputs of each sum gate have the same formal degree, so the convention for empty sums is never invoked below the output (an empty output sum computes $f_n = 0$ with formal degree 0, which is consistent). The circuit computes f_n, and it has at most $V + B + nB + (ns + B) + s\,p(n) + 1$ vertices. □

Remark (unused vertices). The circuit constructed in the proof may contain vertices that no gate reads, for instance a sum $P_t(y_i)$ with $t > k_i$ or a variable on which f does not depend. The model allows this: fan-out is unrestricted, and the formal definition of the model that accompanies [1] does not require a vertex to be read by any gate. The case in which no term remains, so that the output is an empty sum, does not occur in Theorem 3, where $f_n = \mathrm{IMM}_{n,n} \neq 0$.

Table 2 lists the layers of the circuit constructed in the proof, from the output down to the leaves; the bound of the lemma is the sum of the third column.

Layer	Gates	Number	Degree
Output sum	linear combination of the upper products	1	n
Upper products	one for each remaining i and each $\lambda \vdash k_i$	at most $s\,p(n)$	n
Middle sums	$P_t(y_i)$, $1 \le t \le n$, for each remaining i with $J_i^1 \neq \emptyset$	at most ns	t
	unary gates U_j	at most B	1
Lower products	ℓ_j^t, $1 \le t \le n$	at most nB	t
Bottom sums	ℓ_j	at most B	1
Leaves	the variables	V	1

Table 2: The syntactically homogeneous $\Sigma\Pi\Sigma\Pi\Sigma$ circuit for f_n in the proof of Lemma 1, with the formal degree of the gates of each layer. Edges join consecutive rows only (the two kinds of middle sums form one layer), and the total is at most $V + 2B + nB + ns + s\,p(n) + 1$. If no term remains, the output is an empty sum, of formal degree 0.

Lemma 2. *$p(n) \le \exp(\pi\sqrt{2n/3}) < e^{3\sqrt{n}}$ for every $n \ge 1$.*

Proof. For $0 < x < 1$, $p(n)x^n \le \sum_{j\ge 0} p(j)x^j = \prod_{k\ge 1}(1-x^k)^{-1}$ and $\log\prod_k(1-x^k)^{-1} = \sum_{m\ge 1}\frac{1}{m}\frac{x^m}{1-x^m} \le \frac{\pi^2}{6}\frac{x}{1-x}$, since $1 - x^m \ge mx^{m-1}(1-x)$. With $x = e^{-u}$, $u > 0$, and $x/(1-x) = 1/(e^u - 1) \le 1/u$ we get $p(n) \le \exp(\pi^2/(6u) + nu)$; take $u = \pi/\sqrt{6n}$. □

4 Conditional lower bounds and the upper bound

Theorem 3. *Assume Hypothesis A, with its threshold n_0. Let K be a field of characteristic zero. The $\Sigma\Pi\Sigma$ circuits below may have, besides the n^3 entries of the matrices, any finite number of further variables.*

(i) For every $n \ge n_0$, every $\Sigma\Pi\Sigma$ circuit over K computing $\mathrm{IMM}_{n,n}$, with s product gates and B distinct bottom affine forms, satisfies

$$s\,p(n) + ns + (n+2)B + n^3 + 1 \;\ge\; n^{\sqrt{n}/400}.$$

(ii) There is $n_1 \ge 2$, depending only on n_0, such that for every $n \ge n_1$ every such circuit satisfies $s + B \ge n^{\sqrt{n}/400}e^{-4\sqrt{n}} = n^{\sqrt{n}/400 - 4\sqrt{n}/\ln n}$.

Proof. (i) Substitute zero for the variables of the circuit outside the n^3 matrix entries; affine forms remain affine, s and B do not increase, and the circuit still computes $\mathrm{IMM}_{n,n}$. Apply Lemma 1 with the n^3 matrix entries as the variable set, so $V = n^3$, and $f = f_n = \mathrm{IMM}_{n,n}$ (the normalisation step of its proof takes care of forms that have become constant), then Hypothesis A. (ii) By Lemma 2 the left-hand side in (i) is at most $(s+B)(e^{3\sqrt{n}} + n + 2) + n^3 + 1$. Choose $n_1 \ge \max(n_0, 2)$ such that $n^3 + 1 \le \frac{1}{2}n^{\sqrt{n}/400}$ and $2(e^{3\sqrt{n}} + n + 2) \le e^{4\sqrt{n}}$ for $n \ge n_1$. The equality in the statement is $e^{-4\sqrt{n}} = n^{-4\sqrt{n}/\ln n}$, which requires $\ln n \neq 0$, that is $n \ge 2$. □

Remark (leaves). If "gates" in Hypothesis A is read as excluding leaves, which is the stronger reading (see Section 2), then Theorem 3 holds with n^3 deleted from the left-hand side of the inequality in (i), since the circuit of Lemma 1 has n^3 leaves.

Lemma 4. *For $n \geq 2$ and $M = n^2 - n + 1$ there is an $M \times M$ matrix A with entries in the variables of $\mathrm{IMM}_{n,n}$ and $\{0, 1\}$ such that $\operatorname{per} A = \mathrm{IMM}_{n,n}$ and $\det A = (-1)^{n-1}\mathrm{IMM}_{n,n}$.*

Proof. Take the layered graph of $\mathrm{IMM}_{n,n}$ with source and sink identified to one vertex v_0, and add a loop of weight 1 at every other vertex; there are $1+n(n-1)$ vertices, and A is the weighted adjacency matrix. Deleting v_0 leaves only loops as cycles, so each cycle cover is one n-cycle through v_0, corresponding to a path counted by $\mathrm{IMM}_{n,n}$, together with loops; an n-cycle has sign $(-1)^{n-1}$. □

Figure 1 shows the graph of the proof for $n = 3$.

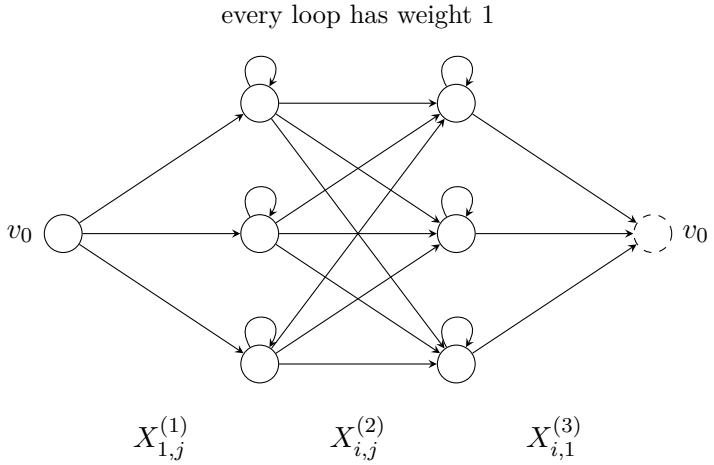


Figure 1: The graph in the proof of Lemma 4 for $n = 3$, with $M = 7$ vertices. The vertex v_0 is drawn twice, as the source on the left and as the sink on the right (dashed); the two are the same vertex. In each of the two inner layers the vertices are numbered $1, 2, 3$ from the top; the edge from v_0 to the jth vertex of the first inner layer has weight $X^{(1)}_{1,j}$, the edge from the ith vertex of the first inner layer to the jth vertex of the second has weight $X^{(2)}_{i,j}$, and the edge from the ith vertex of the second inner layer to v_0 has weight $X^{(3)}_{i,1}$. Each of the six inner vertices has a loop of weight 1.

Corollary 5. *Assume Hypothesis A. There are an absolute constant $c > 0$ and M_0, depending only on n_0, such that over every field of characteristic zero and for every $M \geq M_0$, every $\Sigma\Pi\Sigma$ circuit computing $\det_M$ or per_M has size $s + B \geq M^{cM^{1/4}}$.*

Proof. Let $n = \lfloor \sqrt{M} \rfloor$, so $n^2 - n + 1 \leq M$, and let A' be the block diagonal $M \times M$ matrix with blocks A from Lemma 4 and an identity matrix; then $\operatorname{per} A' = \mathrm{IMM}_{n,n}$ and $\det A' = (-1)^{n-1}\mathrm{IMM}_{n,n}$. Substituting the entries of A' into a $\Sigma\Pi\Sigma$ circuit for per_M or $\det_M$ (any further variables of the circuit are left as they are) gives a $\Sigma\Pi\Sigma$ circuit of the form (1) with no more product gates and no more distinct bottom forms, computing $\mathrm{IMM}_{n,n}$ in the case of the permanent and $(-1)^{n-1}\mathrm{IMM}_{n,n}$ in the case of the determinant. In the latter case, if n is even, replace every coefficient c_i in (1) by $-c_i$: this changes neither s nor B, and the circuit now computes $\mathrm{IMM}_{n,n}$. Apply Theorem 3(ii), using $n \geq \sqrt{M} - 1$; any $c < 1/800$ will do for $M \geq M_0$. □

Remark (the constant). The proof gives the following: for each fixed $c < 1/800$ there is M_0, depending on c and on n_0, such that the bound of Corollary 5 holds with this c for every $M \geq M_0$.

Proposition 6 (Upper bound; [3])**.** *Over every field of characteristic zero, $\mathrm{IMM}_{n,n}$ has $\Sigma\Pi\Sigma$ circuits of size $n^{O(\sqrt{n})}$.*

Proof. By [3, Theorem 1.1], a polynomial over $\mathbb{Q}$ in N variables of degree $d = N^{O(1)}$ computed by an algebraic branching program of size s is computed by a $\Sigma\Pi\Sigma$ circuit of size $2^{O(\sqrt{d \log N \log s})}$. The polynomial $\mathrm{IMM}_{n,n}$ has $N = n^3$, $d = n$ and a branching program with $n^2 - n + 2$ vertices, so it has $\Sigma\Pi\Sigma$ circuits of size $n^{O(\sqrt{n})}$ over $\mathbb{Q}$, hence over every field of characteristic zero. □

Under Hypothesis A the depth-three complexity of $\mathrm{IMM}_{n,n}$ over characteristic zero would therefore be $n^{\Theta(\sqrt{n})}$.

5 Discussion

Range of validity. The bounds are asymptotic. The manuscript [1] gives no value for the threshold n_0 of Hypothesis A: its proof takes n sufficiently large and uses absolute constants that are not computed. Independently of n_0, the bound in Theorem 3(ii) is smaller than 1 unless $\ln n > 1600$, that is $n > e^{1600}$. The inequality in Theorem 3(i) is implied by $s \geq 1$ alone whenever $p(n) \geq n^{\sqrt{n}/400}$; since $p(n) = 2^{\Theta(\sqrt{n})}$ (see [6, Section 23.1]), it carries information only when $\ln n$ exceeds a quantity of order 10^3.

The remark [11] claims explicit thresholds for the depth-five bound: size at least $n^{c\sqrt{n}}$ with $c = 1/400$ for $\ln n \geq 17.6$, with $c = 1/8$ for $\ln n \geq 57.941$, and with every $c < 1/(4\sqrt{2})$ for all large n. These thresholds do not come from the proof in [1] as written but from a modification of it, together with explicit values for the constants of [1, Sections 4 and 5]. The remark is unrefereed, and its claim depends, as Hypothesis A does, on [1, Proposition 4.1]; it is not used here, and n_0 remains unspecified in our statements. If the claim of [11] with $c = 1/8$ were taken as the hypothesis in place of Hypothesis A, the proof of Theorem 3 would give $s + B \geq n^{\sqrt{n}/8} e^{-4\sqrt{n}}$ for $\ln n \geq 58$, which exceeds 1 in that whole range; but then $n > 10^{25}$, and $\mathrm{IMM}_{n,n}$ has more than 10^{75} variables. In either case the bounds say nothing for any n of practical size.

The strength of Hypothesis A. Unconditionally, a depth-three lower bound of the form $n^{\Omega(\sqrt{d})}$ for $\mathrm{IMM}_{n,d}$ over characteristic zero is known for $d = O(\log n)$ [2, 4], and with the loss $2^{-O(d)}$ of [4] it remains superpolynomial up to d of order $\log^2 n$. Hypothesis A asserts much more. Theorem 3 gives one way to test it: $\Sigma\Pi\Sigma$ circuits computing $\mathrm{IMM}_{n,n}$ over a field of characteristic zero of size at most $n^{c\sqrt{n}}$, for a constant $c < 1/400$ and infinitely many n, would refute Hypothesis A.

Characteristic. Characteristic zero is used twice: in the identity expressing e_k through power sums, where characteristic greater than n would suffice as in [2], and in [1].

The determinant. For $\det_M$ the upper bound of [3] is $M^{O(\sqrt{M})}$, so Corollary 5 leaves a gap between $M^{1/4}$ and $M^{1/2}$ in the exponent. The exponent $M^{1/4}$ comes from having the hypothesis only for width equal to degree.

A Formalisation

Part of this note has been formalised in Lean 4 with Mathlib. Table 3 says which statements are covered. The Lean sources are distributed with this note as ancillary files; the accompanying README says how to build them and what each file proves.

Statement	Formalised	Note
Hypothesis A	as a hypothesis	not proved; stated with the definitions accompanying [1]
Lemma 1	in the case used	variable set of $\mathrm{IMM}_{w,w}$, any width w; so $V = w^3$
Lemma 2	yes	
Theorem 3	yes	(i) and (ii), assuming Hypothesis A
Lemma 4	determinant only	by a different proof; the permanent is not covered
Corollary 5	determinant only	with $c = 1/4800$, assuming Hypothesis A
Proposition 6	no	

Table 3: Coverage of the formalisation.

The hypothesis. Hypothesis A enters every formal statement that depends on it as an explicit hypothesis; it is not proved and not assumed as an axiom. It is stated with the definitions of the circuit model, of its size and of $\mathrm{IMM}_{n,n}$ taken verbatim from the Lean development that accompanies [1]; as a formal statement it is the characteristic-zero part of the theorem stated there. The proof given in that development is not imported, and that development and ours have not been combined in one build. The formal proofs use no axioms beyond the three standard ones of Lean.

Not formalised. Lemma 1 for a general variable set; the permanent parts of Lemma 4 and Corollary 5; Proposition 6 and the bound $n^{\Omega(\log n)}$ derived from [4]; the statement that any $c < 1/800$ will do in Corollary 5; the remarks after Lemma 1, Theorem 3 and Corollary 5; and Hypothesis A itself.

Differences from the written proofs. The formal proofs establish the statements and do not follow the written proofs step by step.

- The formal circuit for Lemma 1 is built without the normalisation step.
- In place of the closed form displayed under "Newton's identities" in the proof of Lemma 1, the formal proof shows only that $e_k(y_i)$ is a linear combination, with some coefficients indexed by the partitions λ of k, of the products $\prod_r P_r(y_i)^{\mu_r}$ (by induction on k from Newton's recurrence $ke_k = \sum_{t=1}^{k} (-1)^{t-1} e_{k-t} P_t$, on dividing by k), which gives the same circuit and the same count of vertices with these coefficients in the output gate; the displayed closed form of the coefficients is not checked.
- The formal proof of Lemma 4 for the determinant is by a block factorisation, so the cycle-cover argument given above is not checked by it, and it does not show that the same matrix has permanent $\mathrm{IMM}_{n,n}$.
- The formal Corollary 5 is proved with the constant $c = 1/4800$.

Statement on the use of AI

This note was produced with substantial use of generative AI, and the reader should weigh it accordingly. The author chose the problem area (the permanent versus determinant problem and arithmetic circuit lower bounds), decided to build on the OpenAI mathematics release and to treat its statements as hypotheses, set the procedure by which the work was done and checked, and made the decisions on scope. The observation recorded here was found by the AI system. The

statements, the proofs, the numerical computations, the literature search, and the text of the note were produced by instances of Anthropic's Claude (Claude Opus 5.5), run as autonomous agents under the author's direction. The literature search for a statement of the consequence recorded here was made by these agents on 12 October 2026 and covered the titles of recent arXiv listings in cs.CC and of the ECCC reports of 2026, GitHub (including the forks of the repository of [1]) and the web. A search of this kind does not prove that no such statement exists. The checking was also done by AI: before posting, the note was reviewed internally over several rounds by separate instances of the same model, its citations were compared with the cited sources by further Claude instances (some possibly of a smaller Claude model), and the Lean formalisation described in Appendix A was written and independently read by such instances. The author also obtained comments on the note from GPT-6, a model of a different developer (OpenAI). A later assessment of originality and impact by the same model pointed to the papers [9] and [10], which were then opened and are now cited, and led to the sentence of the Introduction stating that the note contains no new lower-bound technique. The comments and the assessment are both reviews by an AI system, and neither is a verification of the proofs. The note has not been refereed by a human. No AI system is an author; the author takes responsibility for the contents.

References

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